

UNIT 1

PHYSICS AND MEASUREMENTS



PHYSICS:

The word physics is derived from Greek word FUSIS, meaning nature or natural things.

“Physics is defined as that branch of science, which studies natural phenomena in terms of basic laws and physical quantities.”

BRANCHES OF PHYSICS:

The main branches of physics are:

1. Classical or Newtonian physics
2. Modern or Quantum physics

CLASSICAL PHYSICS:

Classical physics believes in singular nature i.e. only the particle nature of matter. It provides macroscopic vision of matter. It is based upon Newton's laws of mechanics and Maxwell's laws of electromagnetism.

Some branches of classical physics are:

1. Mechanics
2. Acoustics
3. Optics
4. Thermodynamics
5. Electromagnetism

MODERN PHYSICS:

Modern physics believes in dual nature of matter i.e. both particle and wave nature of matter. It provides microscopic vision of matter.

Some branches of modern physics are:

1. Atomic physics
2. Nuclear physics
3. Quantum physics
4. Spectroscopy
5. Relativistic physics

SUPPLEMENTARY UNITS:

The units that are used along with base units to form derived units in the International System are called supplementary units.

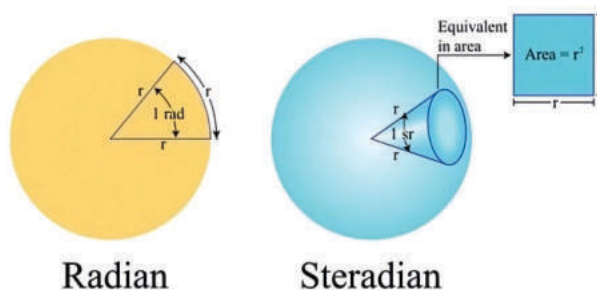
• Radian (rad):

A unit of measurement of angles equal to 57.3° , equivalent to the angle subtended at the center of a circle by an arc equal in length to the radius as shown in figure 1.1(a).

• Steradian (Sr):

The solid angle subtended at the center of a sphere by an area of its surface equal to the square of the radius of that sphere as shown in figure 1.1(b).

Physical Quantities	Supplementary Unit	Symbol	Definition
Plane Angle	Radian	rad	A unit of measurement of angles equal to 57.3° , equivalent to the angle subtended at the center of a circle by an arc equal in length to the radius as shown in figure
Solid Angle	Steradian	Sr	The solid angle subtended at the center of a sphere by an area of its surface equal to the square of the radius of that sphere as shown in figure

**DIMENSIONS:**

It is used to denote the nature of a physical quantity. The symbols L, M, and T are used to specify the dimensions of any physical quantity. The nature of a physical quantity is always expressed as some combinations of the fundamental quantities mass, length, and time.

DIMENSIONS OF BASE/FUNDAMENTAL QUANTITIES:

No.	BASE QUANTITY	SYMBOL OF DIMENSION
1.	Length	[L]
2.	Mass	[M]
3.	Time	[T]
4.	Electric current	[I]
5.	Temperature	[θ]
6.	Luminous intensity	[J]
7.	Amount of substance	[N]

DIMENSIONS OF DERIVED QUANTITIES:

S.No	PHYSICAL QUANTITIES	RELATION	DIMENSIONAL FORMULA	UNITS
1.	Pressure	Force $\div$ Area	$M^1 L^{-1} T^{-2}$	Nm ²
2.	Impulse/Momentum	Force $\times$ time	$M L T^{-1}$	N·s
3.	Power	Work $\div$ Time	$M^1 L^2 T^{-3}$	J/s
4.	Velocity of light	Velocity	$M^0 L^1 T^{-1}$	m/s
5.	Frequency	1 $\div$ Time period	$M^0 L^0 T^{-1}$	s ⁻¹
6.	Velocity gradient	Velocity $\div$ Distance	$M^0 L^0 T^{-1}$	s ⁻¹
7.	Wave length	Velocity $\div$ Frequency	$M^0 L^1 T^0$	m
8.	Gravitational constant (G)	Force $\times$ (Distance) ² $\div$ (mass) ²	$M^{-1} L^3 T^2$	Nm ² /kg ²
9.	Planck's constant (h)	Energy $\div$ Frequency	$M L^2 T^{-1}$	J·s
10.	Strain	Change in length $\div$ Original length	$M^0 L^0 T^0$	No unit

11.	Stress	Force / Area	$M^{-1} L^{-1} T^{-2}$	Nm^{-2}
12.	Rate of flow	Volume / Time	$M^0 L^3 T^{-1}$	m^3/s

USING DIMENSIONS TO DERIVE EQUATIONS OR APPLICATION IN EQUATIONS:

Consider the oscillations of a simple pendulum. We assume that the period of the pendulum [T] depends on following quantities:

- (i) the mass of the pendulum bob [M]
- (ii) the length of string of the pendulum [L],
- (iii) the gravitational acceleration (g) [LT^{-2}]

Therefore, the equation can be written as:

$$T = k m^x l^y g^z \quad g^z = \left(\frac{L}{T^2}\right)^z = \frac{L^z}{T^{2z}}$$

Where x, y and z are unknown powers and k is a dimensionless constant.

The dimensional form above equation is

$$[T] = [M]^x [L]^y [L]^z [T]^{-2z}$$

Equating the indices for M, L and T on both left and right sides of the equation, we get:

$$M: 0 = x$$

$$L: 0 = y + z$$

$$T: 1 = -2z$$

Therefore

$$x = 0, y = \frac{1}{2} \text{ and } z = -\frac{1}{2}$$

The original equation therefore becomes

$$T = k (L/g)^{1/2}$$

Which is what we would expect for a simple pendulum. Dimensionality does not give us the value of the dimensionless constant k which can be shown by other methods to be 2π , so it becomes

$$T = 2\pi \sqrt{\frac{L}{g}}$$

APPLICATIONS OF DIMENSIONS:

Dimensional analysis is a fundamental aspect of measurement and is applied in real-life physics. We make use of dimensional analysis for three prominent reasons:

- To check the consistency of a dimensional equation.
- To derive the relation between physical quantities in physical phenomena
- To change units from one system to another.

HOMOGENEITY PRINCIPLE OF DIMENSIONAL ANALYSIS:

Principle of Homogeneity states that:

“Dimensions of each of the terms of a dimensional equation on both sides should be the same.”

This principle is helpful because it helps us convert the units from one form to another.

ERROR:

“Difference between the measured value and actual value is known as error.”

Types of errors:

1. Systematic Errors
2. Random Errors

1. SYSTEMATIC ERROR:

“Errors that have a clear cause and can be eliminated for the future experiments are known as systematic errors.”

TYPES OF SYSTEMATIC ERROR:

Systematic error is further classified into four types:

1. Instrumental error
2. Observational error
3. Environmental error
4. Theoretical error

I. Instrumental error:

“When the instrument being used does not function properly causing error in the experiment known as instrumental error.”

E.g., such as a scale that reads 2g more than the actual weight of the object, causing the measured values to read too high consistently.

II. Observational error:

“When the scientist (Observer) inaccurately reads a measurement wrong, the error produced is known as observational error.”

E.g., such as when not standing straight-on when reading the volume of a flask, causing the volume to be incorrectly measured.

III. Environmental error:

“When the surrounding environment (such as a lab) causes errors in the experiment is known as environmental error.”

E.g., The scientist cell phone's RF waves cause the Geiger counters to incorrectly display the radiation.

2. Random Errors:

"The errors occur randomly, and sometimes have no source/cause are known as random error."

Types of random error:

1. Observational error.
2. Environmental error.

I. Observational error:

"When the observer makes inconsistent observational mistakes then the error produced is known as observational error."

E.g., Such as not reading the scale correctly and writing down values that are constantly too low or too high.

II. Environmental error:

"When unpredictable changes occur in the environment of the experiment then the error produced, this error is known as environmental error."

E.g., Such as students repeatedly opening and closing the door when the pressure is being measured, causing fluctuations in the reading.

UNCERTAINTY:

"The degree of doubt associated with a measurement, prediction, or estimation is known as uncertainty."

RULES FOR PROPAGATION OF UNCERTAINTY AND ERRORS:

The value after " $\pm$ " is known as error or uncertainty and represents with "A" symbol.

• For Addition and subtraction:

If $x = (A \pm \Delta A)$ and $y = (B \pm \Delta B)$

$x + y = (A + B) \pm (\Delta A + \Delta B)$

$x - y = (A - B) \pm (\Delta A + \Delta B)$

For multiplication and division:

If $x = (A \pm \Delta A)$ and $y = (B \pm \Delta B)$ then,

$$xy = \left\{ (AB) \pm \left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times AB \right\}$$

$$\frac{x}{y} = \left\{ \left(\frac{A}{B} \right) \pm \left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times \frac{A}{B} \right\}$$

SIGNIFICANT FIGURES:

The number of digits which are reasonably reliable are called significant figures.

RULES TO COUNT SIGNIFICANT FIGURES:

The number of digits which are reasonably reliable are called significant figures.

1. Non-zero digits are all significant. E.g., 363 has three significant figures.
2. Zero between non-zero digits are significant. E.g., 5004 has four significant figures.
3. Zero locating the decimal point in numbers less than one is not significant. E.g., 0.062 has only two significant figures.
4. Final zeros to the right of the decimal point are significant figures. E.g., 2.000 has four significant figures.
5. If a number is less than one, then only the zeros that are at the end of the number and zeros that are between non-zero digits are significant. E.g., 0.00420 min has three significant figures.
6. Zero that locates the decimal point in numbers larger than one are not necessary significant. E.g., 400 cm may have one, two, or three significant figures.